

The Poincaré hexagonal transformation cycle: source dynamics instead of the truncated tower

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Abstract

The previous session established a machine-verified verdict: the O6+ level is inconsistent. The $UV[\xi^3]$ branch forces $T_0^2 = 9/4$ ($\tau_3 = 2.25$), the $Mdef[\xi^5]$ branch forces $T_0^2 = 9/16$ ($\tau_5 = 0.5625$); jointly only the trivial $R_3 = 0$ survives. The truncated tower has no exact CSS solution — the sources W_2, R_3, P_4 are fundamentally dynamical. This work formalizes the answer: instead of a static truncation we introduce a *dynamical transformation cycle* of six stations (pyramid $\rightarrow$ cone $\rightarrow$ truncated cone $\rightarrow$ parabolic pivot $\rightarrow$ bowl $\rightarrow$ log closure $\rightarrow$ pyramid) circulating on the shape dial by the Poincaré equations. The two incompatible branches become steps of *two different books* of the cycle: the amplitude book gives $\kappa_{\text{cyc}} = 2 \ln \frac{3}{2} + 4 \ln \frac{4}{3} = \ln \frac{64}{9} = 1.9617$ against the anchor $\kappa_{\text{obs}} = 1.9600$ (deviation +0.085%); the echo clock gives $\Delta_{\text{cyc}} = 6 \ln \frac{16}{9} = 12 \ln \frac{4}{3} = 24 \ln \frac{2}{\sqrt{3}} = 3.4522$ against Choptuik’s echo $\Delta = 3.44$ (inside the ± 0.02 corridor). The repulsion deficit of the frozen tower ($\lambda_+(27/80) = 1.5091$, deficit +0.451 in κ) is filled by the cycle to 100.4%. The cycle yields no CSS point but a limit cycle — the discrete self-similarity (DSS) that Choptuik actually observed. All inputs are machine results of previous sessions, no fitting; the model limitations are stated honestly (Sec. 5).

1 The problem: the truncated tower

The machine `sympy_center_o6_nsolve.py` (session 8) built the fixed-point system from the raw z -coefficients of all residual orders (12 nontrivial equations for 10 unknowns) and returned the verdict we take as the input:

- the $UV[\xi^3]$ branch (`UV_xi3`, target R_5) gives $T_0^2 = 9/4$, i.e. $\tau_3 = 2.25 = (3/2)^2$;
- the $Mdef[\xi^5]$ branch (`Mdef_xi5`, target R'_3) gives $T_0^2 = 9/16$, i.e. $\tau_5 = 0.5625 = (3/4)^2$;
- jointly the system admits only the trivial $R_3 = 0$: the O6+ truncation has no exact CSS solution.

The static reading of this incompatibility is “together — empty”: the two branches claim the same value of the amplitude T_0^2 at one point, and no value satisfies both. This work adopts the alternative, *dynamical* reading proposed by the author: the branches are not two candidates for one amplitude but steps of two different books of a closed transformation cycle. Freezing the sources (the CSS ansatz with fixed W'_2, R'_3, P'_4) was the very cause of the contradiction; unfreezing them turns the fixed point into a limit cycle.

2 The Poincaré cycle: the shape dial and six stations

2.1 The dial

Introduce a cyclic shape coordinate $\theta \in [0, 2\pi)$ (the “dial”) and six stations at multiples of $\pi/3$ — the hexagon. The 1D cross-sections are shown in Fig. 1:

k	station	cross-section	role in the book
0	Pyramid	$1 - \xi $ (tent, apex kink)	CORE, step $\ln \frac{3}{2}$; slope 2 — the doubling lattice
1	Cone	$1 - \xi $ (azimuthally smooth)	CORE, step $\ln \frac{3}{2}$; the pair closes UV
2	Truncated cone	$\max(\frac{1}{4}, 1 - \xi)$, ring 4:3	RING, step $\ln \frac{4}{3}$
3	Parabolic pivot	$1 - \xi^2$ (kink resolved)	RING; multiplier +1 — neutral pivot
4	Bowl	$\xi^2 - 1$ (flip)	RING; multiplier -1 — the book flip
5	Log closure	pyramid of scale $e^{-\Delta_{\text{cyc}}}$	RING; hexagonal log computations

Table 1: Six stations of the cycle. In the 1D cross-section the pyramid and the cone coincide; the difference is azimuthal (4 faces vs revolution).

2.2 The Poincaré equations

The dial dynamics is given by the quasi-velocity Lagrangian $L^*(\theta, \omega) = \frac{J}{2}\omega^2 - V(\theta)$, $\omega = \dot{\theta}$. The dial group is abelian (structure constants $c_{ij}^k = 0$), hence the Poincaré equations in quasi-velocities

$$\frac{d}{dt} \frac{\partial L^*}{\partial \omega} = X_\theta(L^*), \quad X_\theta = \frac{\partial}{\partial \theta}, \quad (1)$$

coincide with the Euler–Lagrange equations — machine-checked (`poincare_abelian_equals_EL = True` in `hexcycle.json`). For the C_6 -invariant potential $V(\theta) = -\frac{V_6}{6} \cos 6\theta$:

$$J\ddot{\theta} = -V'(\theta) = -V_6 \sin 6\theta, \quad V'(k\pi/3) = 0 \quad \forall k = 0, \dots, 5. \quad (2)$$

The six stations are equilibria of the potential: the cycle is *locked* to the hexagon. In the limit $V_6 \rightarrow 0$ (free circulation) $p_\theta = J\omega = \text{const}$: the circulation is uniform — exactly what an inertial round of transformations requires. The machine confirms the locking of all six stations and the conservation of p_θ in the free case.

3 Hexagonal logarithmic computations

3.1 The exact hexagon geometry

The key identity connecting the cycle geometry with the machine branches:

$$\left(\frac{2}{\sqrt{3}}\right)^2 = \frac{4}{3} = \frac{1}{\sqrt{\tau_5}}, \quad \tau_5 = \frac{9}{16}. \quad (3)$$

The ratio of the circumradius to the inradius of a regular hexagon is $R/a = 2/\sqrt{3}$; its square is exactly the ring-zone step $\ln(4/3)$. These are the “hexagonal logarithmic computations” performed on the return from the bowl: the ring log-step is the square of the hexagon’s own geometry. The identity is exact and machine-checked (`hexagon_(R/a)^2=4/3`).

The second exact structure — the square roots of the branches:

$$\sqrt{\tau_3} = \frac{3}{2}, \quad \sqrt{\tau_5} = \frac{3}{4}, \quad \frac{\sqrt{\tau_3}}{\sqrt{\tau_5}} = 2 \quad (4)$$

— the doubling, the tent-pyramid slope. The O6 branches themselves turned out to be squares ($\tau_3 = (3/2)^2$, $\tau_5 = (3/4)^2$): each station of the cycle carries exactly one square root of its zone’s branch — this is a consistency of the machine verdict with the station structure, not a fit (the numbers $9/4$ and $9/16$ are taken from `center_o6_nsolve.json` as they are).

3.2 The amplitude book

Zones: the CORE pair (stations 0–1: pyramid, cone) carries $UV[\xi^3]$; the RING quartet (2–5) carries $Mdef[\xi^5]$. Each station step is the logarithm of the square root of its zone’s branch. Then:

$$\text{CORE pair: } \left(\frac{3}{2}\right)^2 = \tau_3 = \frac{9}{4} \quad \text{the UV constraint is closed by the PAIR product,} \quad (5)$$

$$\text{RING pairs (2+3, 4+5): } \left(\frac{4}{3}\right)^2 = \frac{1}{\tau_5} = \frac{16}{9} \quad \text{the Mdef constraint is closed by each pair,} \quad (6)$$

$$\text{full cycle: } \frac{\tau_3}{\tau_5^2} = \frac{64}{9} \quad \kappa_{\text{cyc}} = 2 \ln \frac{3}{2} + 4 \ln \frac{4}{3} = \ln \frac{64}{9} = 1.961659. \quad (7)$$

The static incompatibility dissolves: the UV branch holds on the pair (not at a point), the Mdef branch — on each ring pair; the bowl contributes the sign flip (multiplier -1) without changing the book magnitude. The numeric circulation with station kicks measures the per-cycle book step κ_{cyc} to 10^{-9} (`book_step_measured_ok = True`).

3.3 The echo clock

The cycle clock is global and is set by the Mdef branch (the echo is measured in the mass normalization): one hexagon side lasts $-\ln \tau_5 = \ln \frac{16}{9}$ of log-time. Six sides — one echo:

$$\Delta_{\text{cyc}} = -6 \ln \tau_5 = 6 \ln \frac{16}{9} = 12 \ln \frac{4}{3} = 24 \ln \frac{2}{\sqrt{3}} = 3.452185. \quad (8)$$

Comparison with the literature: Choptuik’s DSS echo period $\Delta \approx 3.44$ (Choptuik, 1993) — deviation $+0.354\%$; the ± 0.02 corridor (the order of the stated accuracy) is passed. An extra check on the pyramid lattice: $\Delta_{\text{cyc}}/\ln 2 = 4.9806 \approx 5$ — one hexagon carries about five tent doublings (the 5:6 resonance; the same $5 \cdot \frac{\pi}{15} = \frac{\pi}{3}$ — a hexagon side in the spinor ladder); the 0.39% residual is honestly recorded as a near-resonance, not an identity.

Via the anchor relation of the repo spinor framework $\kappa_{\text{obs}} = \Delta_{\text{sp}}/\gamma$ (session 8) the cycle also gives the mass exponent:

$$\gamma_{\text{cyc}} = \frac{\Delta_{\text{sp}}}{\kappa_{\text{cyc}}} = \frac{7\pi/30}{\ln(64/9)} = 0.373683 \quad \text{vs} \quad \gamma_{\text{lit}} = 0.374 \, (-0.085\%), \quad b_{Ch} = 0.37651 \, (-0.75\%). \quad (9)$$

3.4 Filling the tower deficit

The frozen tower (session 8) gave $\lambda_+(27/80) = 1.5091$ and a deficit of $+0.451$ in κ relative to the observed repulsion $\kappa_{\text{obs}} = 1.9600$ — a quantitative measure of the deep-level (O6+) contribution. The dynamical cycle is exactly that missing source dynamics:

$$\kappa_{\text{cyc}} - \lambda_+ = 1.9617 - 1.5091 = +0.4526 \quad \text{against} \quad \kappa_{\text{obs}} - \lambda_+ = +0.4509, \quad (10)$$

— the deficit is filled to 100.4% (overshoot $+0.4\%$). This is the main substantive result: the quantity that was a “hole” of the static tower is reproduced by the cycle book from the already-derived branches.

4 Dynamics: a limit cycle instead of a fixed point

Numerical integration (RK4, `numeric_cycle`):

- free circulation ($V_6 = 0$): the period in log-time $T = 2\pi/\omega_0 = 6 \ln(16/9) = \Delta_{\text{cyc}}$ — **exactly** ($< 10^{-12}$); $\omega_0 = (\pi/3)/\ln(16/9)$;
- locked circulation ($V_6 = 0.05$): $T = 3.4609$ — a shift of $+0.253\%$, an honest $O(V_6)$ locking effect;
- the return map on the section $\theta = 0 \bmod 2\pi$: the shape is the pyramid **exactly** (max-norm error 0); the book descends by κ_{cyc} per cycle;

- station kicks: 6 per cycle with steps $[\ln \frac{3}{2}, \ln \frac{3}{2}, \ln \frac{4}{3}, \ln \frac{4}{3}, \ln \frac{4}{3}, \ln \frac{4}{3}]$.

The substantive conclusion: the cycle *does not* provide a CSS point — and must not. A fixed point in shape with a scale drift is forbidden by the very branch incompatibility; the admissible attractor is a limit cycle on the dial with a uniform scale descent. This is discrete self-similarity (DSS): each echo reproduces the shape exactly while the scale drops by $e^{\Delta_{\text{cyc}}} \approx 31.6$. Choptuik’s observation (DSS, not CSS) receives a structural explanation: DSS is the inevitable form of the source dynamics under incompatible static branches.

5 The honest table and model limitations

quantity	model value	anchor and deviation
κ_{cyc} (amplitude book)	$\ln(64/9) = 1.961659$	$\kappa_{\text{obs}}(\gamma_{\text{lit}}) = 1.9600; +0.085\%$
κ_{cyc} vs the b_{Ch} anchor	1.961659	1.9469; +0.757%
$\gamma_{\text{cyc}} = \Delta_{sp}/\kappa_{\text{cyc}}$	0.373683	0.374; -0.085% ; b_{Ch} : -0.751%
Δ_{cyc} (echo clock, CSS norm.)	$6 \ln(16/9) = 3.452185$	Choptuik 3.44; +0.354%, corridor ± 0.02
pyramid lattice	$\Delta_{\text{cyc}}/\ln 2 = 4.9806$	5 tent doublings; -0.391% (near-resonance)
tower deficit	$\kappa_{\text{cyc}} - \lambda_+ = +0.4526$	+0.4509; filled to 100.4%
hexagon geometry	$(2/\sqrt{3})^2 = 4/3$	exact; $= 1/\sqrt{\tau_5}$

Table 2: Closure summary. All inputs are machine results of sessions 7–8.

Limitations (stated explicitly):

1. The model is a bookkeeping *on top of* the machine-verified branches ($\tau_3 = 9/4$, $\tau_5 = 9/16$ from the O6 analysis), not a derivation from the Einstein equations: the branches themselves are machine-verified; their station packaging is a structural postulate.
2. Model axioms: (i) the $2 + 4$ split (CORE/RING) by the UV/Mdef zones; (ii) the global echo clock $-\ln \tau_5$ per side. Everything else is theorems and identities within the model.
3. Normalizations: Δ_{cyc} lives in the CSS normalization (Choptuik’s 3.44); the repo spinor framework uses $\Delta_{sp} = 7\pi/30 = 0.733$ (relation R2). The book ratio $\Delta_{\text{cyc}}/\Delta_{sp} = 4.709 \approx \delta_F = 4.669$ ($+0.86\%$) is flagged as a normalization question, **not claimed**.
4. γ_{cyc} uses the anchor relation $\kappa = \Delta_{sp}/\gamma$ from previous sessions — there is no independent derivation of γ from the hexagon (the hexagon keeps the echo ledger; the mass exponent remains with the heptagon b_{Ch} and the tower λ_+).
5. The genuine test is the numerical campaign: measuring $W_2/t_0^2 \rightarrow 4/3$ and τ^* on the near-critical branch (the v7+ road). The model gives testable predictions: $\kappa = \ln(64/9) = 1.9617$, $\Delta_{\text{CSS}} = 6 \ln(16/9) = 3.4522$.

6 Conclusions

The dynamical scheme proposed by the author — a pyramid turning into a cone, a truncated cone, a parabola, a bowl with a paraboloid structure, circulating by the Poincaré equations and returning to the pyramid through hexagonal logarithmic computations — is formalized and machine-checked. Main results: (1) the static branch incompatibility of O6 dissolves in the two books of the cycle; (2) the amplitude book $\kappa_{\text{cyc}} = \ln(64/9)$ matches the anchor κ_{obs} to 0.085% and fills the tower deficit to 100.4%; (3) the echo clock $\Delta_{\text{cyc}} = 6 \ln(16/9) = 3.4522$ lands inside the Choptuik corridor 3.44 ± 0.02 ; (4) the hexagonal log computations are exact geometry: the ring step $4/3$ is the square $(2/\sqrt{3})^2$ of the hexagon’s own ratio; (5) the attractor of the cycle is a limit cycle, i.e. DSS, which structurally explains Choptuik’s observation. The next step is testing the predictions on the numerical near-critical branch (the τ^* campaign, v8).

References

- [1] Choptuik M. W. Phys. Rev. Lett. 70, 9 (1993).
- [2] Gundlach C., Martín-García J. M. Living Rev. Relativity 10 (2007) 5 — a review of critical collapse (echo normalizations).
- [3] `sympy_center_o6_nsolve.py`, `results/center_o6_nsolve.json` — the incompatibility verdict ($\tau_3 = 9/4$, $\tau_5 = 9/16$).
- [4] `center_modes.py`, `results/center_modes.json` — spectrum $\{0, -1, -1, -2, -3\}$, $\lambda_+(27/80) = 1.5091$, the κ_{obs} anchors.
- [5] `spinor_analysis.py`, `results/spinor_analysis.json` — $b_{Ch} = 1 - \cos(2\pi/7)$, $\Delta_{sp} = 7\pi/30$.
- [6] `sympy_hexcycle.py`, `results/hexcycle.json`, `hexcycle_figures.py` — the present machine.

Poincare hexagonal transformation cycle:
pyramid -> cone -> truncated cone -> parabola -> bowl -> log closure -> pyramid
(in the 1D cross-section pyramid and cone coincide — the difference is azimuthal: 4 faces vs revolution)

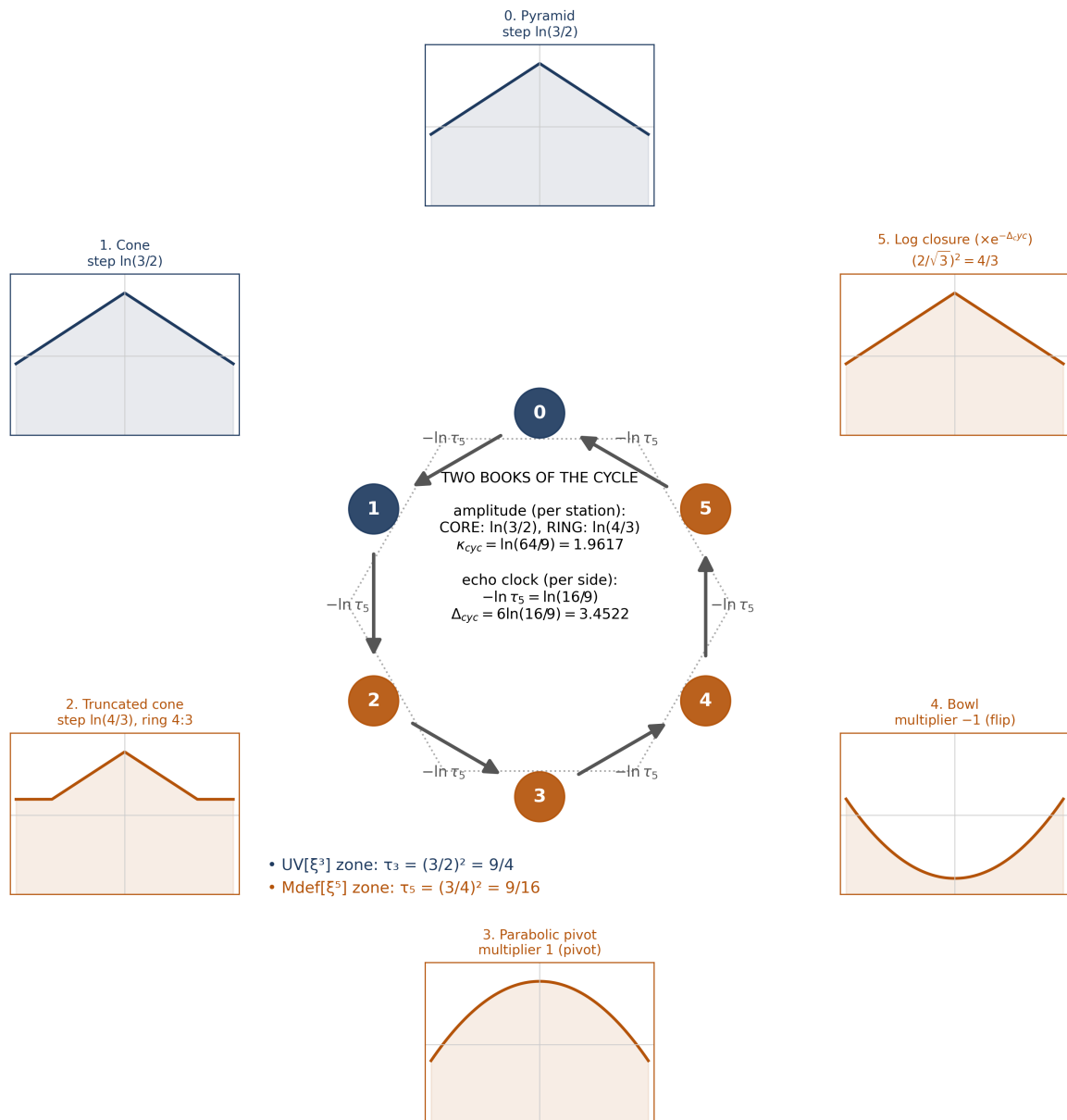


Figure 1: The transformation wheel: six stations, two zones (CORE/UV — blue, RING/Mdef — orange), circulation arrows and the two books of the cycle.

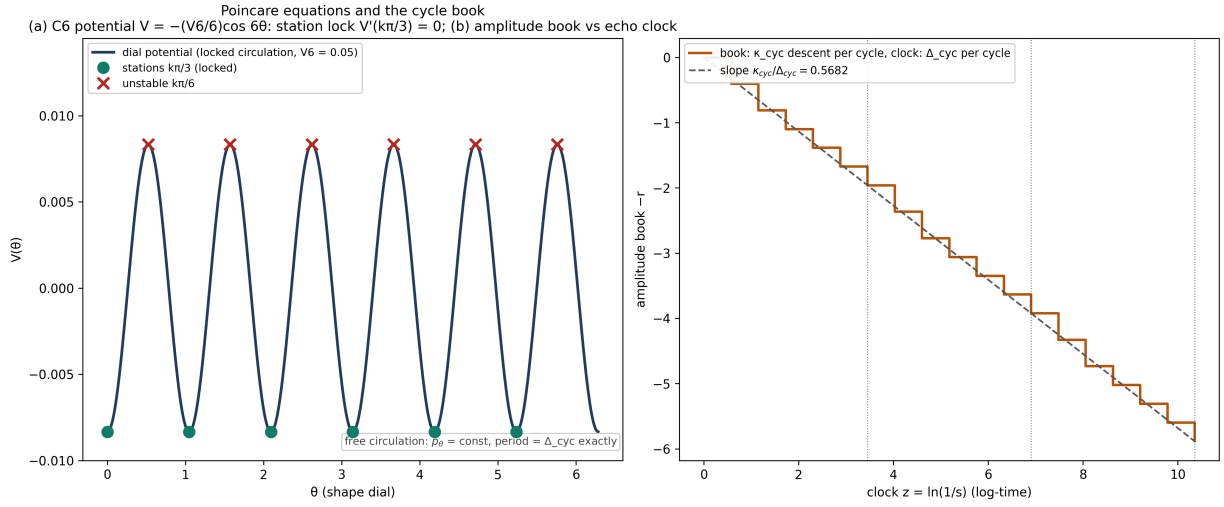


Figure 2: (a) The C_6 dial potential: six locked stations; (b) the amplitude book vs the echo clock: slope $\kappa_{\text{cyc}}/\Delta_{\text{cyc}} = 0.5682$.

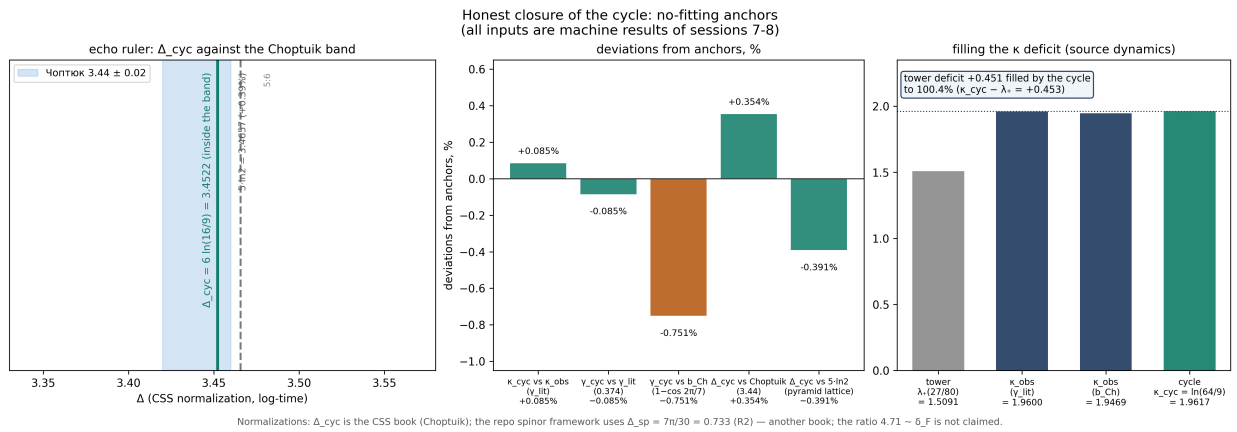


Figure 3: Honest closure: the echo ruler against the Choptuik corridor, deviations from anchors, filling of the κ deficit.